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India SMART UTILITY Week 2026

10 – 14 March 2026, New Delhi

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DEPARTMENT OF SCIENCE & TECHNOLOGY

AIDA
Collaboration | Innovation | Excellence

Session : NEW TECHNOLOGIES: AI–ML–VR–AR AND ROBOTICS APPLICATIONS FOR UTILITIES

Transforming Power Distribution Through Intelligent Technologies

Presented By

Sh Brajesh Kumar, COO, BSES Yamuna Power Limited

DISCOM PROFILE

Joint Ventures

सत्यमेव जयते

Government of the National Capital Territory of Delhi

Joint Venture

BYPL is a joint venture between Reliance Infrastructure Limited and Govt of NCT of Delhi with a 51%: 49% share holding

License Period

25 Years

License Area

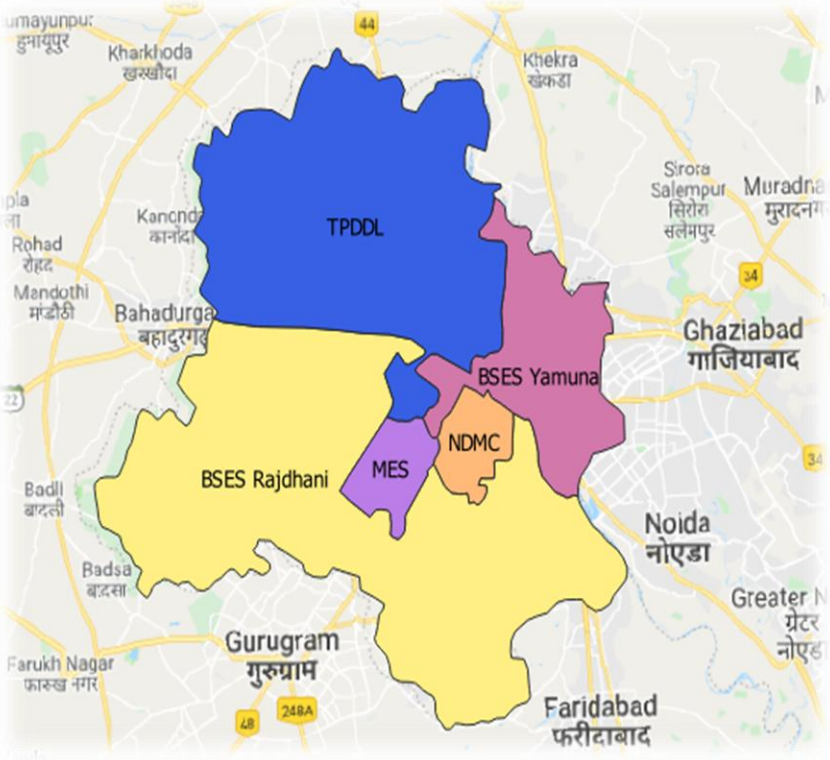
160 sq km in the eastern and Central parts of the National Capital

Customer base

~21 Lacs

System Reliability

99.97 %



BSES Yamuna Power Limited (BYPL) has been the frontrunner in implementing power distribution reforms in the capital city of Delhi and is acknowledged for its quality, reliable power supply and customer friendly practices.

One of the most Successful Public Private Partnership. An ISO 9001:2015 ISO 14001:2015 & BS OHSAS 45001:2018 Certified Organization

BYPL is a joint venture between Reliance Infrastructure Limited and Govt of NCT of Delhi with a 51%: 49% share holding. The company started operations on July 1, 2002 post the unbundling of the erstwhile Delhi Vidyut Board (DVB).

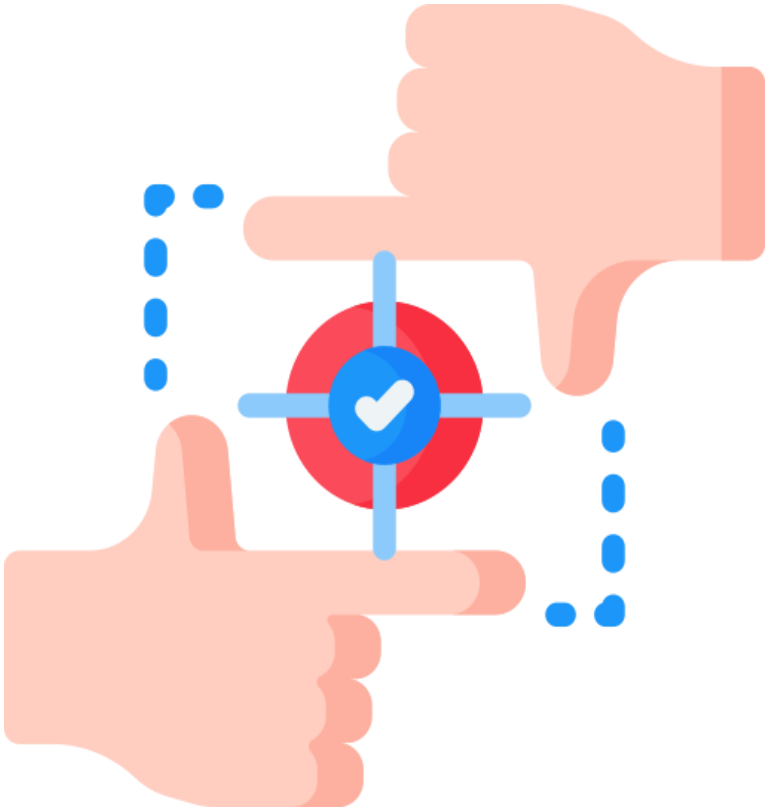
TRANSFORMATION JOURNEY

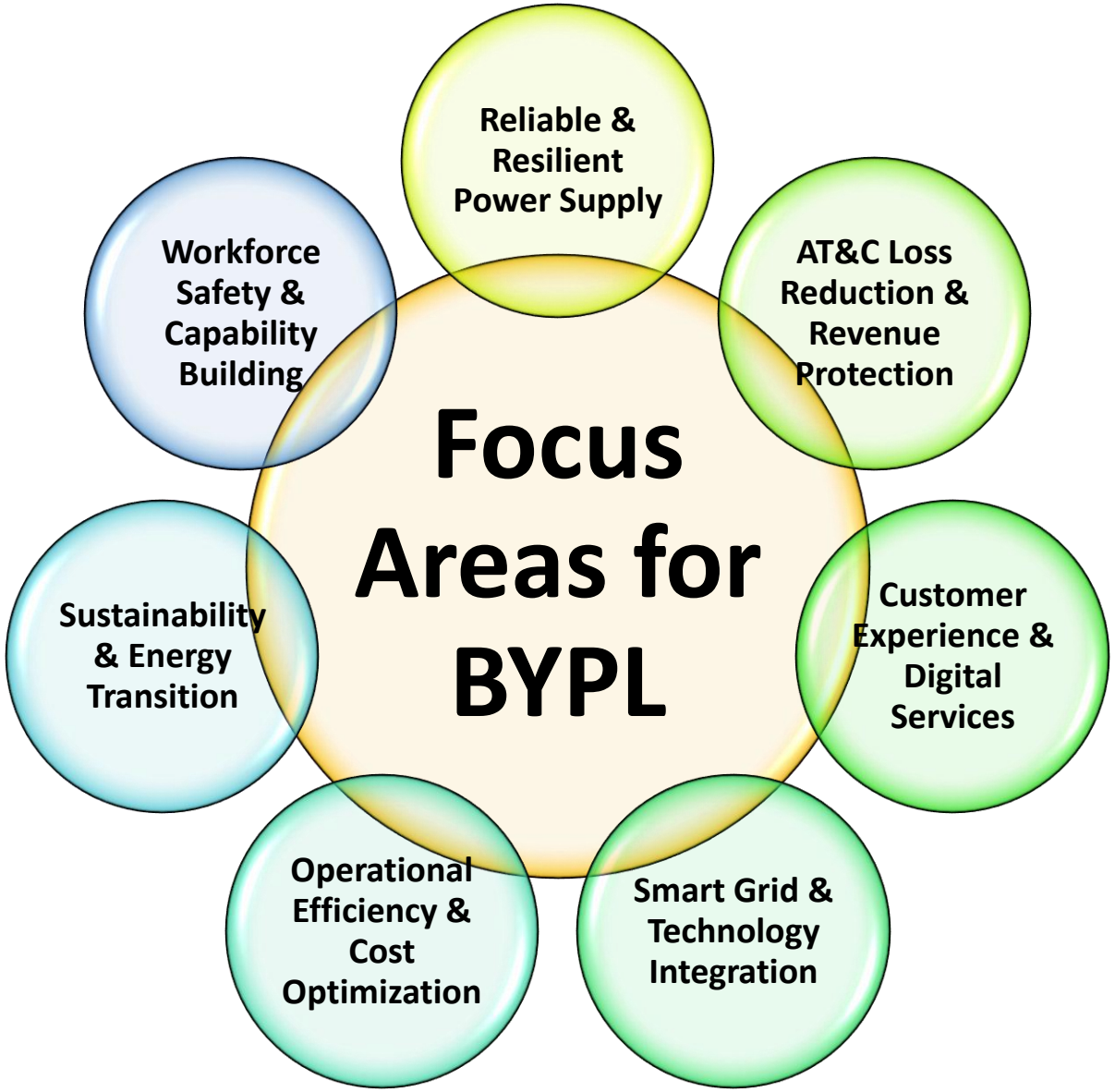
	Then- FY 03	Now- FY 25
Consumer Count	7.5 lacs	20.37 Lacs
Peak Demand	801 MW	1882 MW
AT&C Losses	63%	6.02%
System Reliability	~70%	99.97%
Inaccurate / provisional billing	> 20%	<1%
Transformer Failure Rate	15%	0.66%

BYPL Efforts Include

Manpower	Technology	Digitisation	Network Improvement	Metering & Billing
Skilled and dedicated manpower deputed	New Technologies Introduced	Digitization of manual work done in various departments	Routine preventive maintenance and network enhancement	Quality of meter reading and billing improved

PRIME FOCUS AREAS





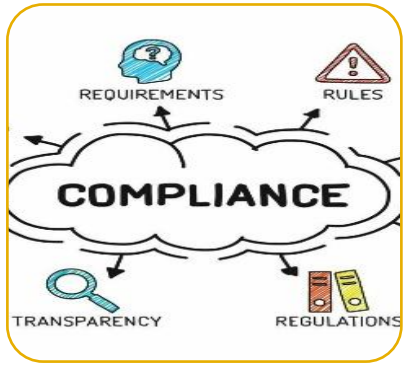
Why Emerging Technologies Are Critical for BYPL

Rapidly Growing Urban Electricity Demand
Increasing population density, urbanization, and rising per-capita consumption are driving higher and more dynamic load patterns across East & Central Delhi.

Expansion of Rooftop Solar & Net Metering
Growth in distributed renewable energy is creating bidirectional power flows, requiring smarter grid management, forecasting, and real-time monitoring.

EV Charging Ecosystem Growth
Accelerating adoption of electric vehicles is introducing new peak loads and localized demand clusters, necessitating predictive load planning and infrastructure readiness.

Extreme Weather & Climate Resilience
Heatwaves, heavy rainfall, and climate variability demand stronger grid resilience, faster fault detection, and proactive asset management.



Stricter Regulatory Performance Benchmarks (SAIDI/SAIFI)
Maintaining and improving reliability indices requires data-driven operations, automation, and rapid outage response mechanisms.

Digital & Emerging Technologies in BYPL

AI-Driven Solutions by BYPL Voice Bot

19122 (24x7 Helpline)

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Area_Power_Outage	• When a customer reports an area-wide power outage, the system fetches details through the API. If the outage is already identified as a known issue, the bot informs the customer accordingly and shares the estimated TAT over the call.
Individual_Power_Outage	• If a customer reports a power issue specific to their residence, RMN and CA validation is completed first. Once validated, the complaint is registered and the complaint number along with the TAT is communicated to the customer.
Get_Complaint_Status	• When a customer calls to check the status of an existing complaint, RMN and CA numbers are validated. If the complaint exists, the customer is updated with the latest progress status.
Wrongly_Complaint_Closed	• If a complaint has been closed without resolution, the system validates RMN and CA details. The bot then confirms whether power has been restored and records the customer's input. After validating all mandatory information, a new complaint is registered and the complaint number is shared.
Telephony Dispositions / API Failures	• In cases such as no answer, API failure, signal issues, or null responses where validation cannot be completed, the call is directly transferred to a live agent for further handling.

AI-Driven Intelligent Document Processing for New Connections

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Overview

Deployment of **NLP-based Artificial Intelligence integrated with OCR** to automate document verification in the new connection process.

OCR digitizes and extracts textual data, while NLP models perform intelligent interpretation, contextual validation, and compliance analysis.

Key Challenges

- Manual processing bottlenecks
- Resource-intensive verification (ownership, address, notarization, authentication)
- Compliance complexity across diverse document formats
- Days-long approval timelines impacting customer satisfaction
- High quality-control overhead for regulatory consistency

AI Platform Solution – Intelligent Document Processing

- LLMs + Google Vision API for intelligent document parsing
- Contextual address extraction & automated verification
- Automated ownership, notarization & relationship validation
- Real-time regulatory compliance mapping

Impact Delivered

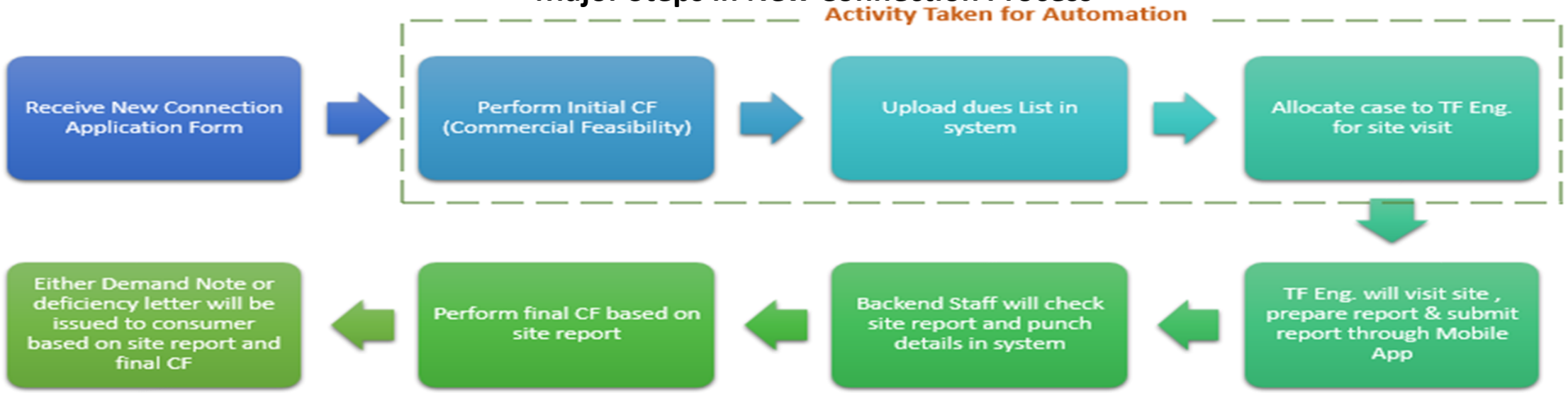
- 80% reduction in processing time (Days → Hours)
- Labor cost optimization & workforce role enhancement
- Improved accuracy, compliance, and customer experience

AI-Powered CF Verification in New Connection Process

Process Overview

This process involves identification of outstanding dues against the applied address during the new electricity connection request. The objective is to validate commercial feasibility before approval of the connection.

Major Steps in New Connection Process



Reason for Identifying the Proposed Steps

- Repetitive nature of the activity
- Time-consuming manual verification process

AI/ML-Enabled Solution – Key Benefits

- Accelerates new connection processing
- 24×7 automated operations (RPA-enabled)
- Improves accuracy and decision-making
- Reduces human error
- Optimizes workforce utilization
- Enhances overall customer satisfaction

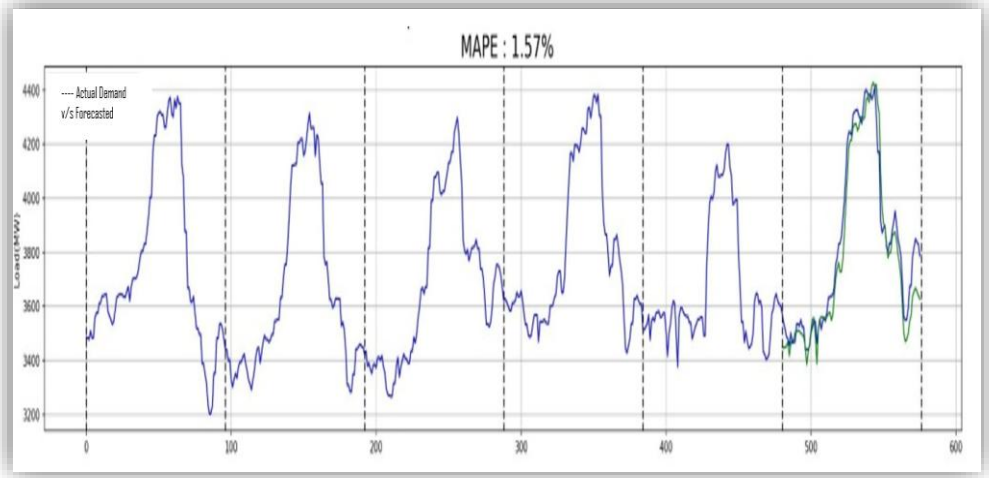
AI/ML Based Demand Forecasting

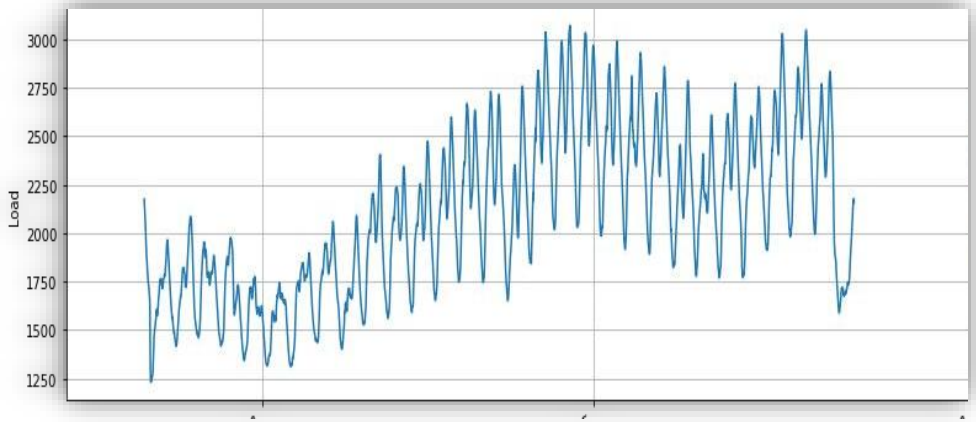
Overview

- Utilizes advanced **Machine Learning** models (Ensemble & Gradient Boosting) for high-precision demand prediction
- Continuously evolves using real-time feedback, weather anomalies & consumption behavior changes
- Automatically retrains using latest SCADA demand data
- Integrates forecasting outputs with power procurement, dispatch & cost optimization decisions

Key Features

- **Multi-Horizon Forecasting:**
  - Day-ahead forecasting (daily for next day)
  - Intra-day & real-time forecasting
  - 15-minute block intraday forecast
- **Dynamic Model Retraining:**
  - Self-learning mechanism adapting to seasonal & load variations

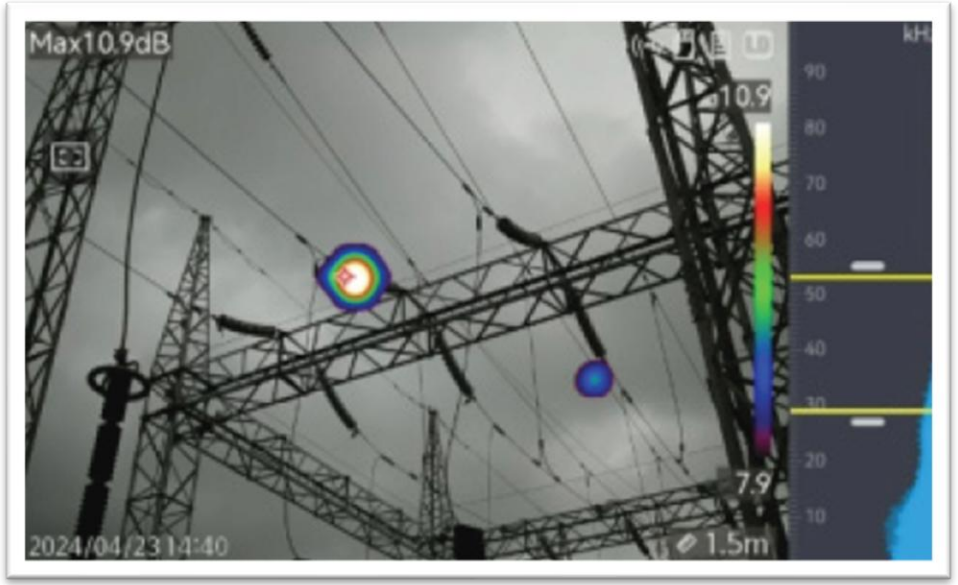


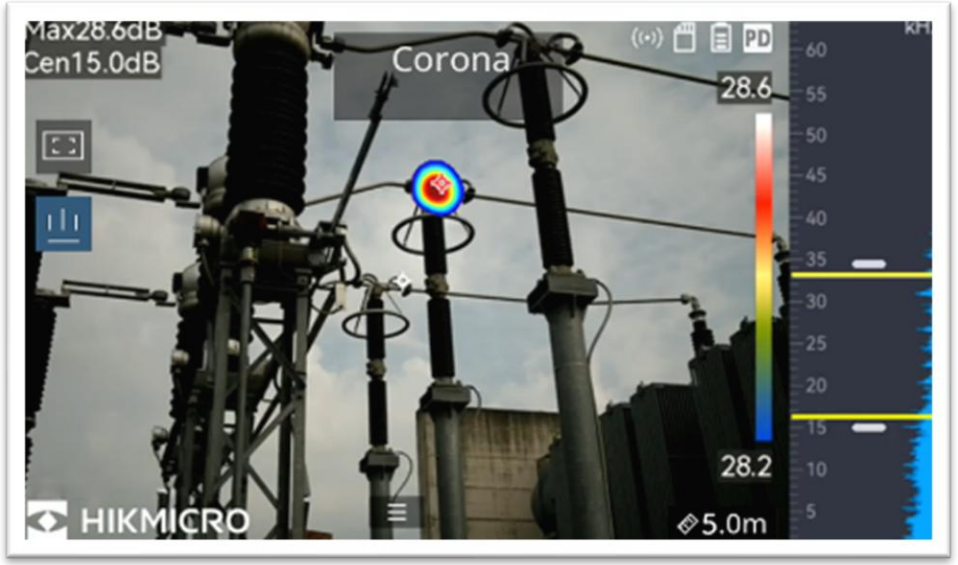


Online Partial Discharge Camera with Acoustic Analyser

Project Name: *Online Partial Discharge Camera with Acoustic Analyser for Overhead Electrical Assets*

For BSES Yamuna Power Limited (BYPL), the PD Measurement Camera integrates multiple sensitive microphones with an LED screen for real-time detection and visualization of partial discharge in overhead assets. This advanced acoustic monitoring system enables early fault identification, improves reliability, and supports predictive maintenance, enhancing network performance and reducing unexpected outages.





E-Surveillance of Transmission Lines & Towers through Drone

Purpose

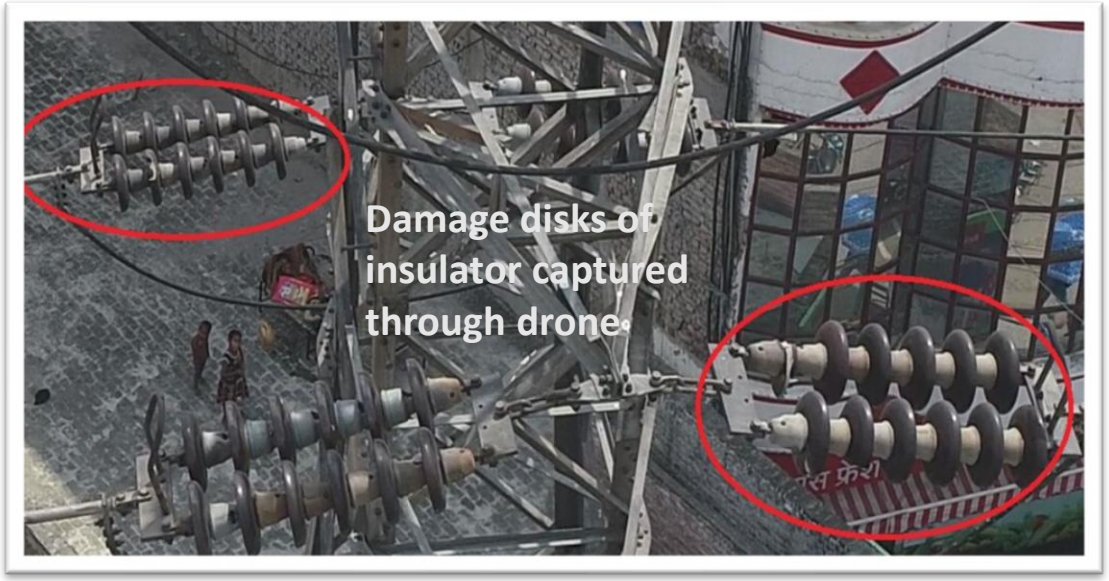
- Reduction in EHV tripping
- Improvement in transmission reliability

Need

- 30–40% breakdowns due to transient faults & external factors
- Ground patrolling limitations:
  - No 360° inspection view
  - Limited accessibility in difficult terrains

Drone Inspection – Key Advantages

- Complete 360° view of lines & accessories
- Access to difficult / inaccessible terrains
- Identification of hot spots (beyond handheld device capability)
- Visual condition assessment of insulators, jumpers & fittings



Aerial view captured through Drone

FUTURE ROADMAP

AI / ML Roadmap

- AI-powered **virtual assistants**
- Smart meter data analytics for **theft pattern detection**
- AI-driven **renewable integration forecasting**
- IoT sensors (temp, oil, vibration) → ML models to predict failures / schedule condition-based maintenance, reducing unplanned outages and replacement costs.
- **Digital Twin Deployment :** Develop and deploy a Digital Twin of the 11kV and LT distribution networks using GIS, SCADA, IoT, and AI/ML to create a real-time virtual grid that enables predictive maintenance, scenario testing for DERs (Distributed Energy Resource) and EV integration, dynamic outage management, and customer impact forecasting, starting with a pilot rollout for high-demand areas.

VR / AR Roadmap

- VR safety & skills training for linemen and O&M staff
- Virtual substation walkthrough training modules
- AR for field operations / remote expert assistance

Robotics Roadmap (primarily drones; other robotics as emerging)

- Aerial inspection with drones - AI-assisted image analytics
- Ground robotics & automation

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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

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